



# Ada GÖRGÜN

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## Research Interests

Computer vision, representation learning, generative models, image-video generation, explainability

## Education

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 07/2024 – Present | <b>Doctor of Philosophy in Computer Science</b><br><i>Supervisors: Prof. Dr. Bernt Schiele, Dr. Jonas Fischer</i>                                                                                                                                                                                                                                                                 | <b>Universität des Saarlandes</b>      |
| 10/2020 – 08/2023 | <b>M.S. in Electrical and Electronics Engineering</b><br><i>Specialization Area: Signal Processing   CGPA: 4.00 / 4.00</i><br><b>Thesis</b>  : Interpreting Convolutional Blocks as Feature Embedding by Template Matching for Image Recognition<br><i>Supervisor: Prof. Dr. A. Aydın Alatan</i> | <b>Middle East Technical Uni. EEE</b>  |
| 10/2015 – 07/2020 | <b>B.S. in Electrical and Electronics Engineering</b><br><i>Specialization Area: Signal Processing and Biomedical Engineering</i><br><i>CGPA: 3.71 / 4.00 Graduated as a High Honor Student</i>                                                                                                                                                                                   | <b>Middle East Technical Uni. EEE</b>  |

## Work Experience

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| 01/2024 – Present | <b>PhD Student</b><br>Researching on representation learning and the explainability of deep networks in the context of computer vision.                                                                                                                                                                                                                                        | <b>Max-Planck-Institute for Informatics</b>  |
| 09/2020 – 12/2023 | <b>Researcher</b><br>-Designing and testing face detection algorithms using <b>Python</b> .<br>-Designed an enhanced SSA simulator for generating realistic space images using <b>Python</b> .<br>-Designed frameworks for detecting and segmenting ships, detecting resident space objects (RSOs) from high-resolution optical images using <b>Python</b> and <b>MATLAB</b> . | <b>METU Center for Image Analysis</b>      |
| 10/2018 – 09/2020 | <b>Undergraduate Student Researcher</b><br>Developed an algorithm for semantic segmentation by uniting the concepts of fully-connected networks and convolutional neural networks with super-pixels using <b>Python</b> .                                                                                                                                                      | <b>METU Center for Image Analysis</b>      |
| 07/2019 – 09/2019 | <b>Intern</b><br>Developed an algorithm for radar target classification on micro-doppler signatures of drones and birds using  LSTM in <b>MATLAB</b> .                                                                                                                                      | <b>ASELSAN, Turkey</b>                     |
| 06/2018 – 08/2018 | <b>Intern</b><br>Developed a foundation on deep learning and implemented semantic segmentation using  U-NET in <b>Python</b> .                                                                                                                                                            | <b>Mobilus-Invidyo, Turkey</b>             |

## Publications

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| 09/2025 | <b>Temporal Concept Dynamics in Diffusion Models via Prompt-Conditioned Interventions Paper</b> <br>Author(s): <b>Ada Görgün</b> , Fawaz Sammani, Nikos Deligiannis, Bernt Schiele, and Jonas Fischer<br>Accepted as a conference paper at the International Conference on Learning Representations, (ICLR 2026) |
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| 07/2025 | <b>VITAL: More Understandable Feature Visualization through Distribution Alignment and Relevant Information Flow</b><br>Author(s): <b>Ada Görgün</b> , Bernt Schiele, and Jonas Fischer<br>Accepted as a conference paper at the International Conference on Computer Vision, (ICCV 2025) | Paper  |
| 08/2023 | <b>Knowledge Distillation Layer that Lets the Student Decide</b><br>Author(s): <b>Ada Görgün</b> , Yeti Ziya Gürbüz, and A. Aydın Alatan<br>Accepted as a conference paper at the British Machine Vision Conference 2023 (BMVC 2023)                                                      | Paper  |
| 10/2022 | <b>Feature Embedding by Template Matching as a ResNet Block</b><br>Author(s): <b>Ada Görgün</b> , Yeti Ziya Gürbüz, and A. Aydın Alatan<br>Accepted as a conference paper at the British Machine Vision Conference 2022 (BMVC 2022)                                                       | Paper  |

## Relevant Projects

**Pattern Recognition:** Compared two methods for ship detection from satellite images, one using  deep-learning, another one using  graph-cut.

**Machine Vision:** Compared two methods for semantic segmentation, one using super-pixels joined with deep learning, another one using  GrabCut.

**Adaptive Signal Processing:** Focused on optimum filtering theory and implemented various adaptive filters including LMS,  $\epsilon$ -NLMS, APA, RLS, Leaky LMS, Signed-error LMS for comparison using **MATLAB**. Implemented a mouse tracker with Kalman Filter using **C** and **LabVIEW**.

**Optimization:** Implemented various optimization algorithms including Gradient Descent Method, Newton Method, Davidon-Fletcher-Powell method, line search algorithms and a project using the enhanced version of the Salp Swarm Algorithm (Available on  Github)

## Honors and Awards

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| 2020        | <b>Bülent Kerim Altay Award</b><br><i>An award given to students who has 4.0 GPA.</i>                                           | <i>Bülent Kerim Altay Awards Website</i>      |
| 2016 – 2020 | <b>High Honor Student</b><br><i>Semester high honor student 6 times. A title given to students with a GPA greater than 3.5.</i> | <i>METU Rules and Regulations Article-28</i>  |

## Certificates & Other Skills

**Certificates:** IELTS Academic 8 / 9

**Simulation & Design:** LT Spice, Quartus II, Verilog, LabVIEW, Keil  $\mu$ Vision

**Utilities:**  $\LaTeX$ (Experienced), GitHub (Experienced), Microsoft Office applications (Experienced), Ubuntu (Experienced)

## Languages

**Natural:** Turkish (Native Speaker), English (Advanced)

**Programming:** Python, MATLAB, C++, C

**Machine Learning Libraries:** PyTorch, TensorFlow, Hugging Face, Scikit-Learn

**Computer Vision Libraries:** OpenCV, PIL

## Hobbies & Other Interests

Playing Piano, Playing Chess, Playing Tennis, Swimming

## References

Available upon request.